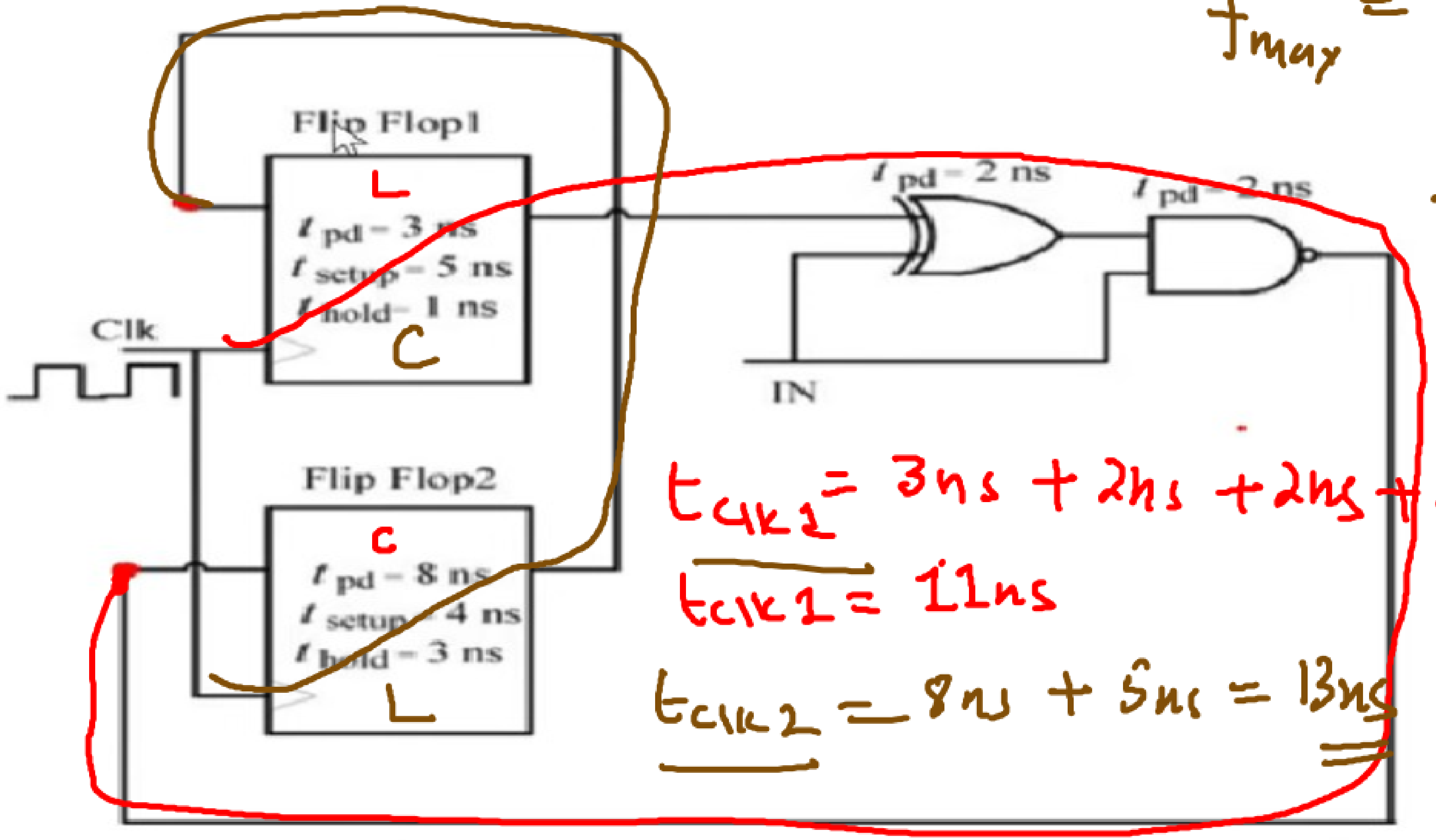
For the components in the sequential circuit shown below, t_{pd} is the propagation delay, t_{setup} is the setup time, and t_{hold} is the hold time. The maximum clock frequency **(rounded off to the nearest integer)**, at which the given circuit can operate reliably, is ___77___ MHz.

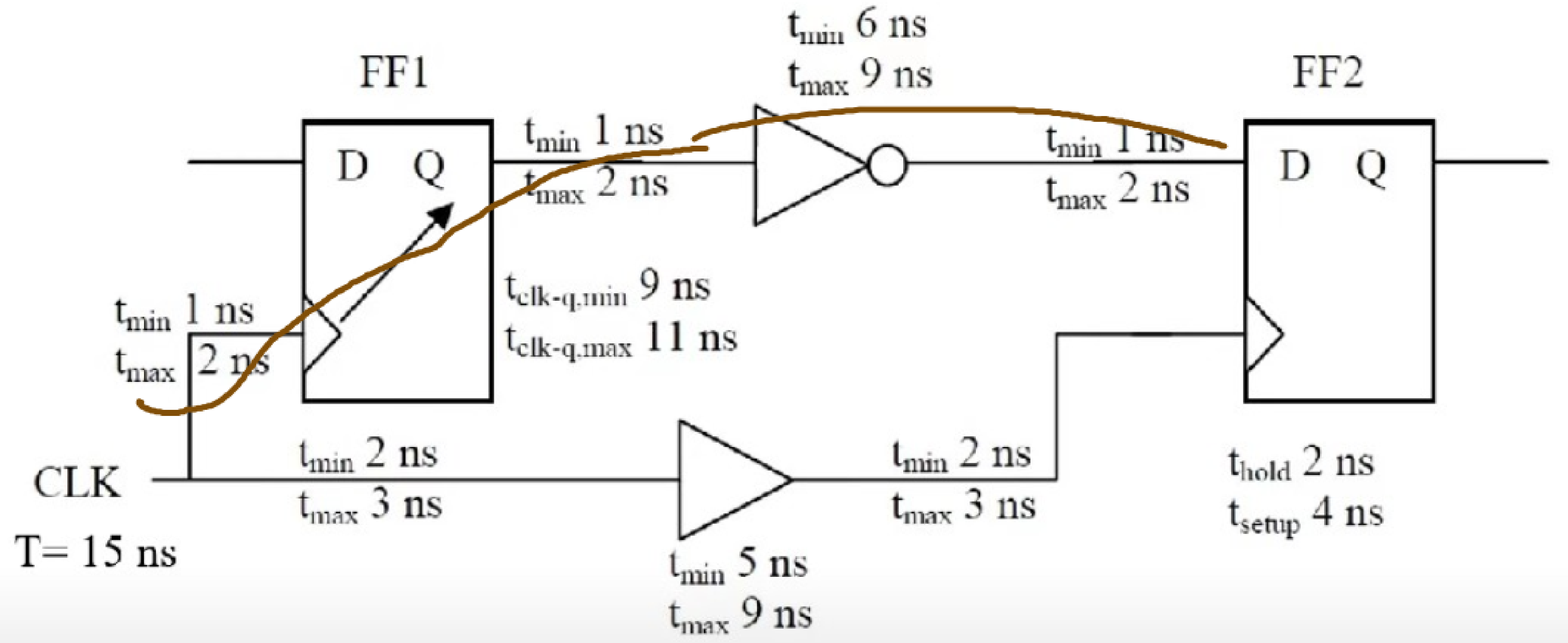


$t_{clk1} = 3ns + 2ns + 2ns + 4ns$

$t_{clk1} = 11ns$

$t_{clk2} = 8ns + 5ns = 13ns$

$$f_{max} = \frac{1}{13 \times 10^{-9}} = \frac{1000}{13 \times 10^{-6}} = 76.9\ MHz$$

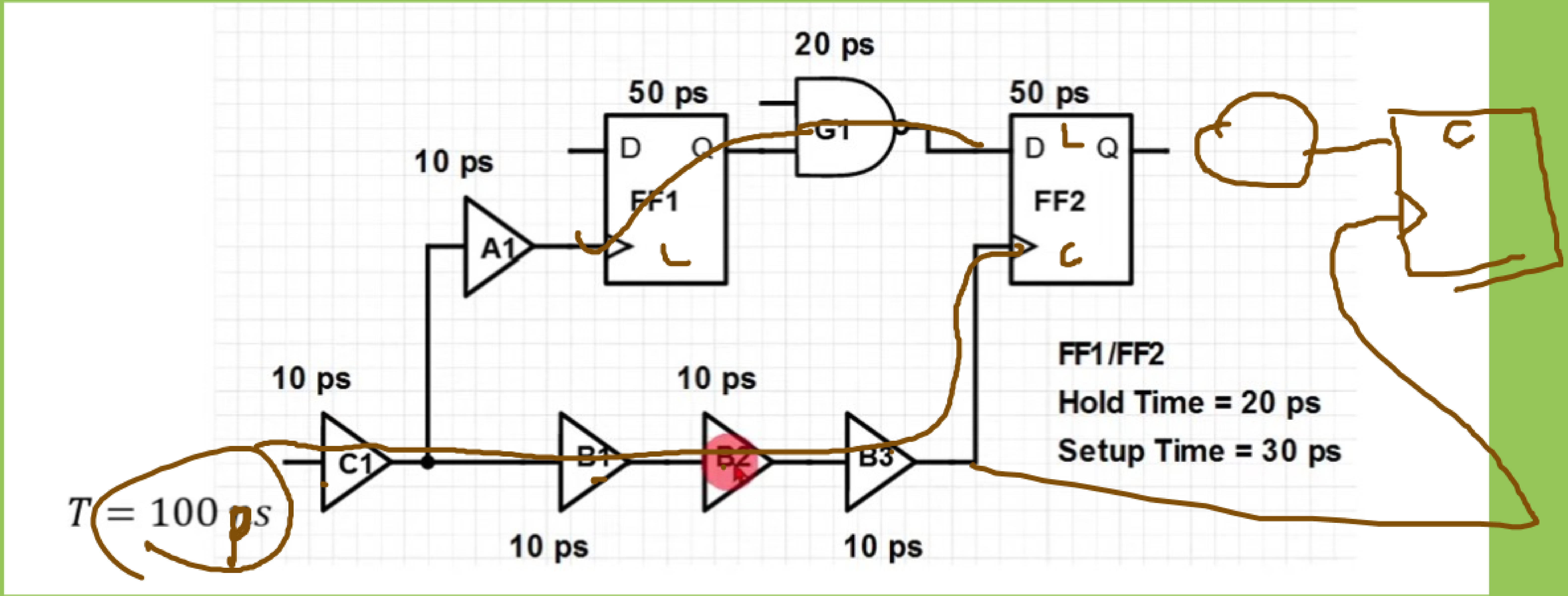


1ns + 9ns + 1ns + 6ns + 1ns ≥ 3ns + 9ns + 3ns + 2ns

Hold slack = 18n − 17ns = +1ns, Hold violation not occured.

Setup slack = 24ns − 30ns
= −6ns

Setup violation is occured.



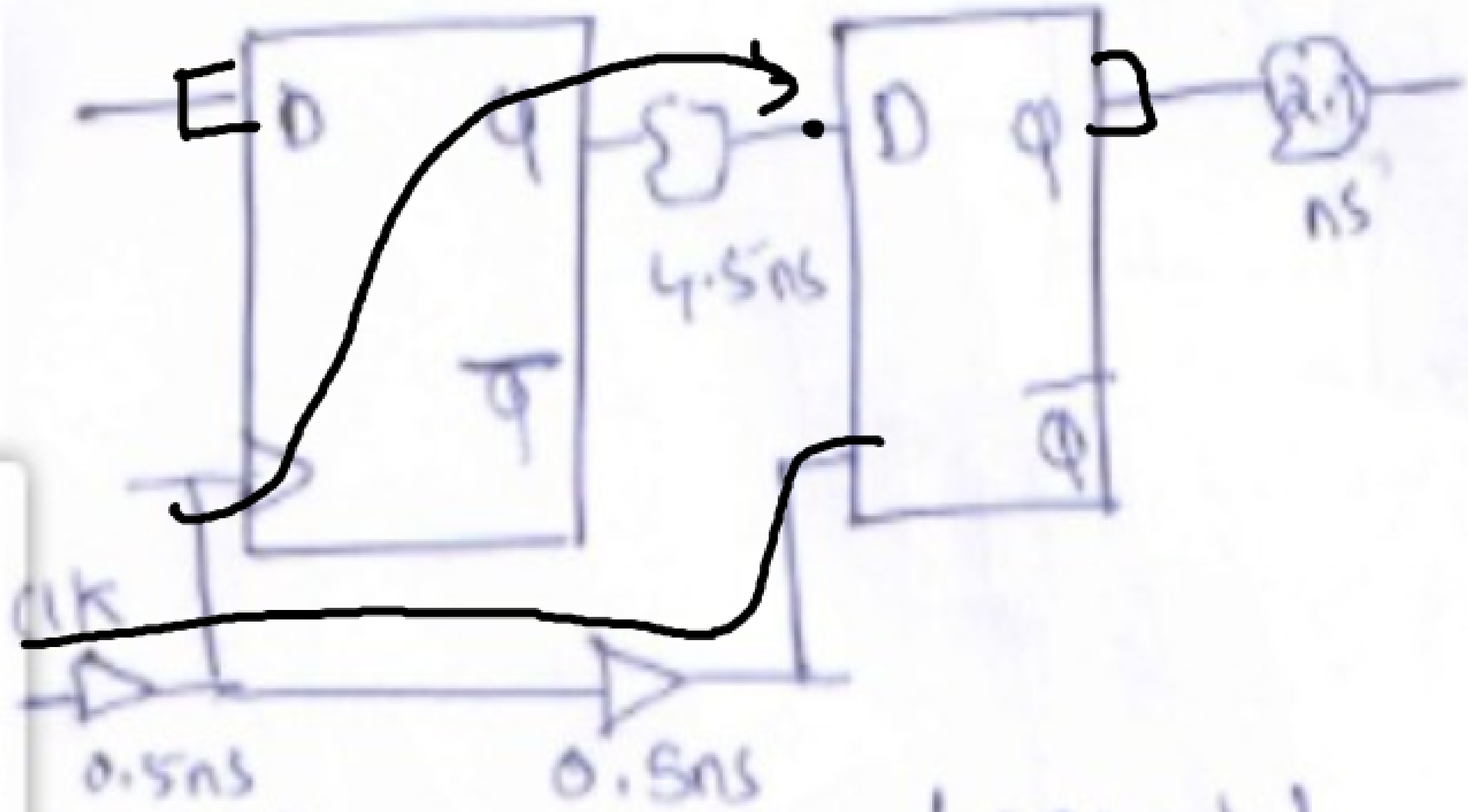
$$10ps + 10ps + 50ps + 20ps + 30ps \leq 100 + 40$$

$$120 \leq 140$$

$$90ps > 60ps$$

No set violation

8) Calculate operating frequency.



→ Pad to reg
→ reg to reg
→ reg to pad.

$t_{cq} = 1ns$
$t_{st} = 0.5ns$
$t_{hld} = 0.4ns$

$t_{cq} = 1.1$
$t_{st} = 0.6$
$t_{hld} = 0.4$

$T_{clk} =$

$T_{clk} + 1ns = 6.6ns$

$T_{clk} = 6.6 - 1ns$
$= 5.6ns$

$\frac{1000}{5.6 \times 10^{-6}}$

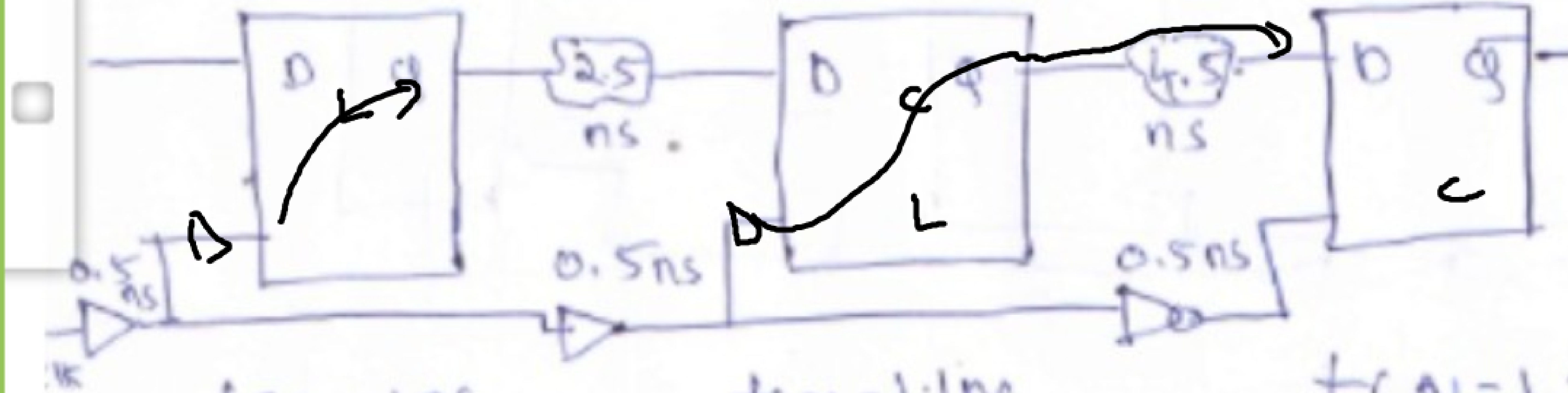
178.5 MHz

9)

$1ns + 2.5 + 0.6 - 1ns = T_{clk_1} = 3.1ns$

$1.1ns + 4.5ns + 0.7 - 1.5$

$t_{clk_2} = 4.8ns$

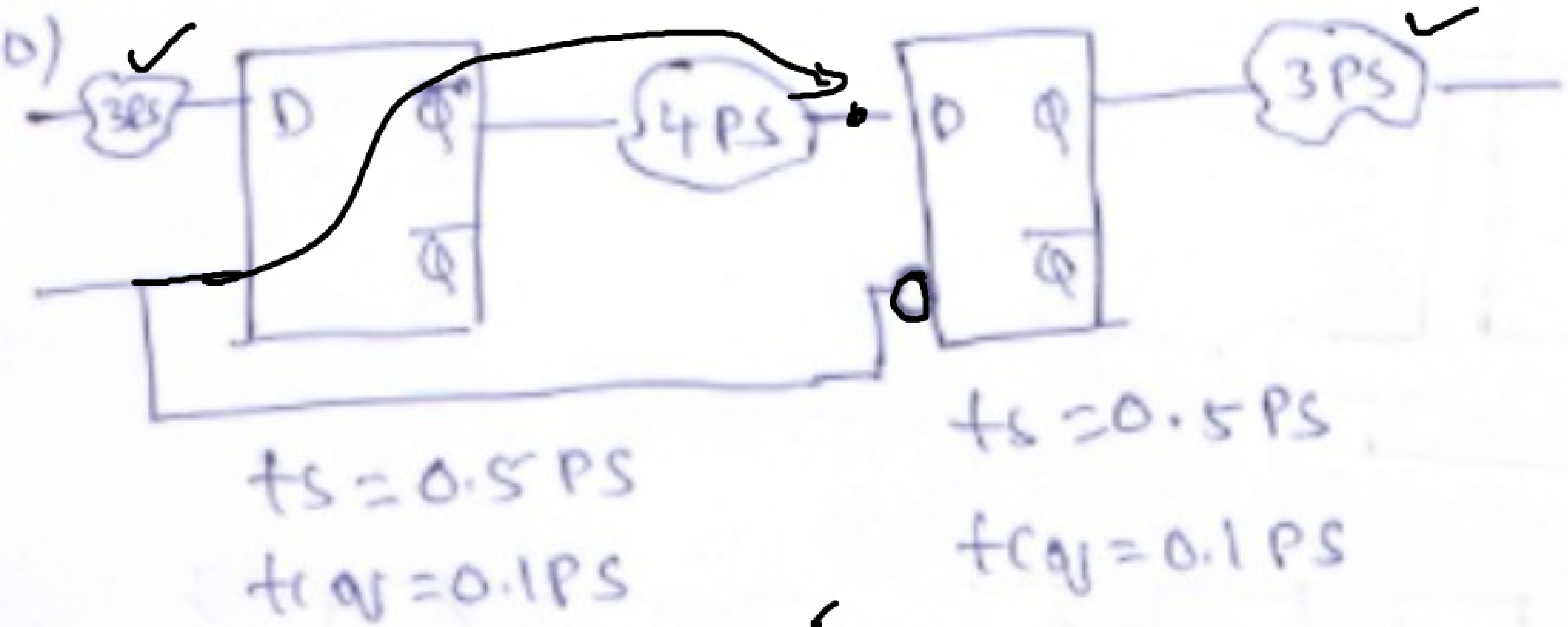


$f_{max} = \frac{1000}{4.8 \times 10^{-6}}$

$= 208.3 MHz$

$t_{cq} = 1ns$	$t_{cq} = 1.1ns$	$t_{cq} = 1.2ns$
$t_{st} = 0.5ns$	$t_{st} = 0.6ns$	$t_{st} = 0.7ns$
$t_{hld} = 0.4ns$	$t_{hld} = 0.4$	$t_{hld} = 0.4ns$

(10)



$0.1 + 4ps + 0.5$

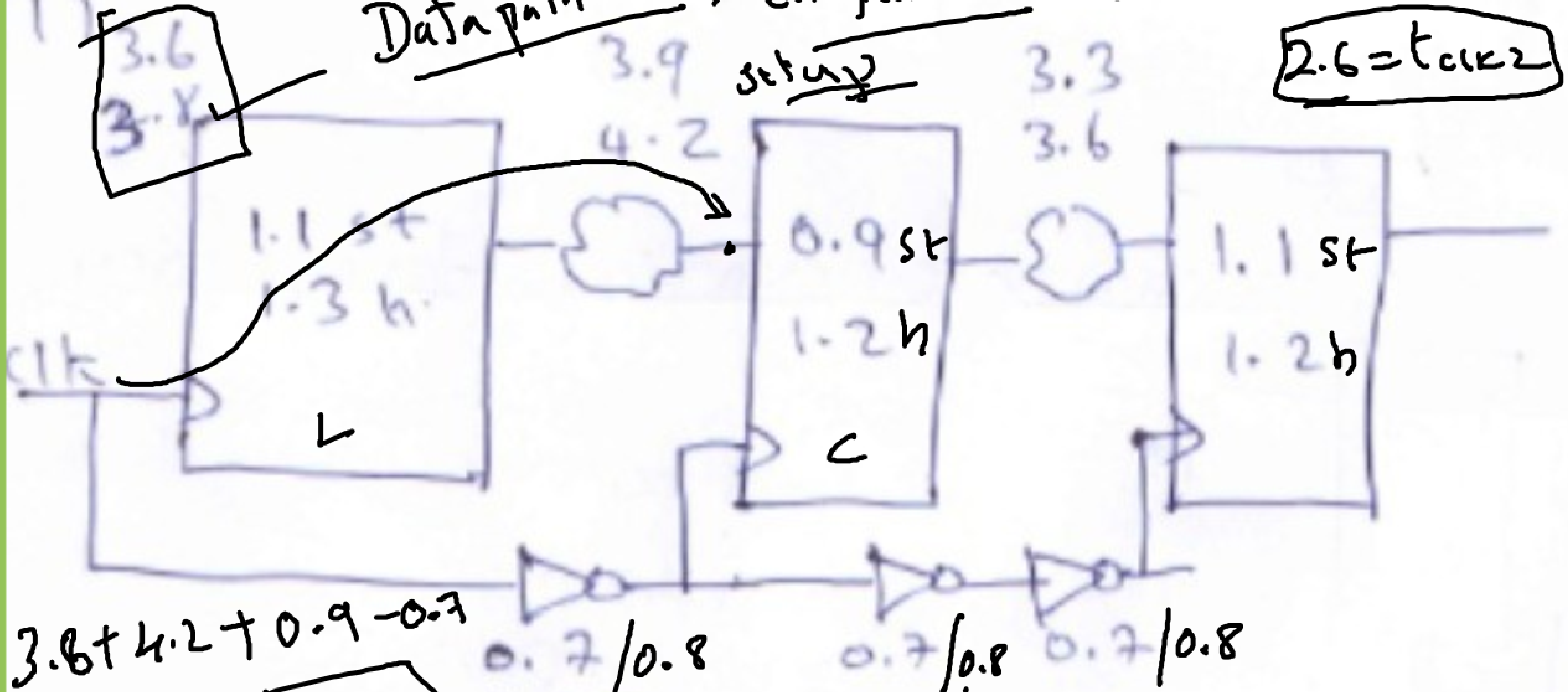
$\underline{4.6ps}$

Datapath max, Clk path, min

$3.6 + 1.1 - 2.1 = t_{clk2}$

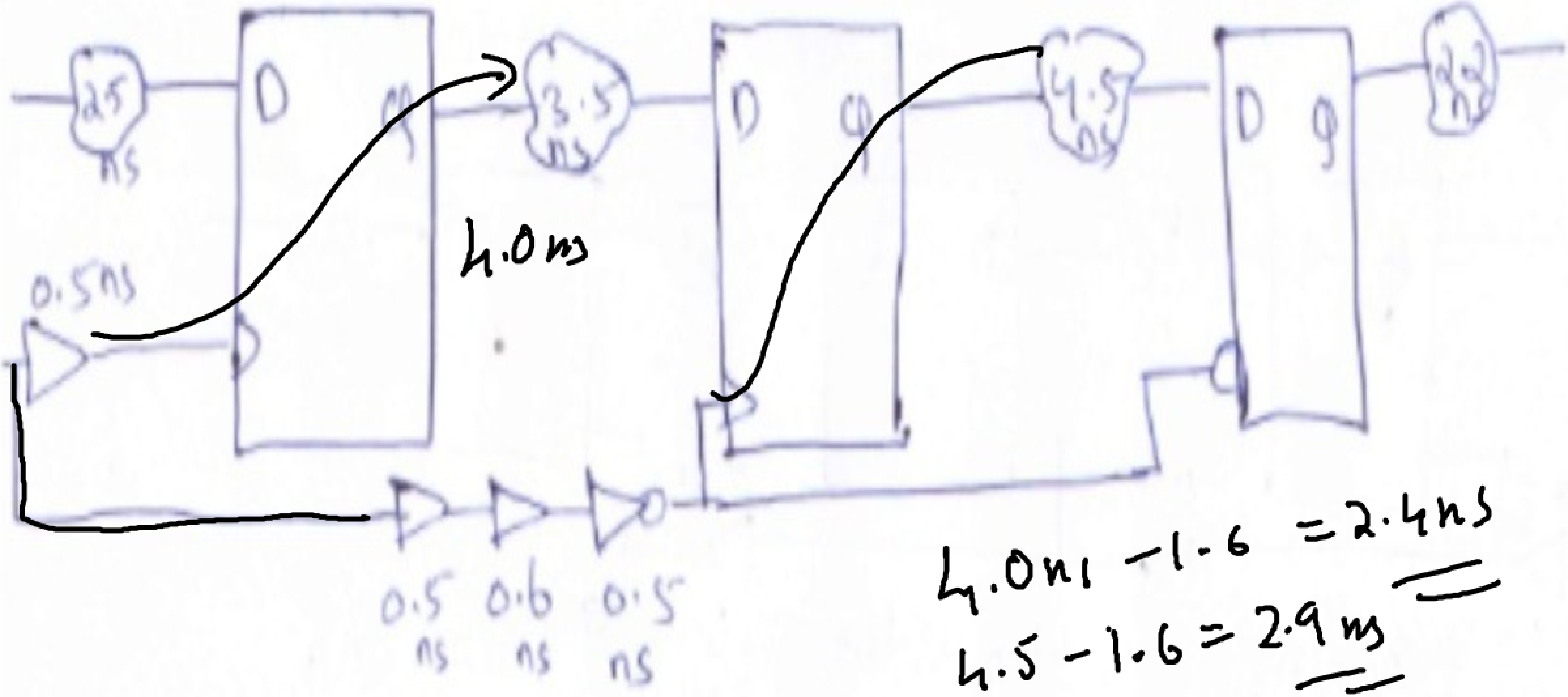
setup

$\boxed{2.6 = t_{clk2}}$



$3.8 + 4.2 + 0.9 - 0.7$

$\boxed{8.2 = t_{clk1}}$


2.5 ns
D
Q
3.5 ns
D
Q
4.5 ns
D
Q
2.2 ns
0.5ns
4.0ns
0.5 ns
0.6 ns
0.5 ns
4.0ns − 1.6 = 2.4ns
4.5 − 1.6 = 2.9ns